

Divanshu Sharma

AI Engineer

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EDUCATION

University of Mumbai | Bachelor of Engineering, Computer Science | June 2025 | CGPA: 8/10.0

- Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Distributed Systems, System Design

EXPERIENCE

Uniiq.ai | Founding Engineer | May 2026 - Present

- Led the technical turnaround of an AWS-deployed Generative AI onboarding platform, resolving 40+ critical vulnerabilities in inherited legacy code.
- Engineered LLM security protocols, building prompt and response sanitization layers to thwart prompt injection attacks, resolving data leaks, and implementing user ID-based rate limiting.
- Optimized full-stack architecture, improving the web performance score from 73 to 93, slashing Largest Contentful Paint (LCP) from 3.3s to 0.7s, and reducing interaction latency (INP) to 130ms.

Yale University | ML Researcher | Oct 2025 - Mar 2026

- Advised by Prof. Haris Smajlović & Prof. Claus Horn to develop a secure, open-source deep learning framework from first principles (Python/NumPy) for privacy-preserving collaborative AI training.
- Architected a compiler-centric Multi-Party Computation (MPC) system, merging core cryptographic components directly into the main secure repository (PR #42).
- Designed and trained a custom CNN with strided convolutions, achieving 88.08% test accuracy on the ChestMNIST benchmark following rigorous failure-mode analysis.

WorldQuant BRAIN | Quantitative Research Consultant | Nov 2022 - July 2025

- Built and automated 15+ high-frequency alpha signal generation workflows using Python and WorldQuant Simulation, entirely replacing manual backtesting processes.
- Engineered models that achieved an average Sharpe Ratio of 1.8, driving automated quantitative trading logic.

PROJECTS & OPEN-SOURCE

Graph Data Explorer AI | [GitHub](#) | [Link](#) | Python, LangChain, Qdrant, FastAPI, React

- Built an end-to-end data pipeline transforming 21,393 transactional records across 19 relational tables into a 765-node/877-edge knowledge graph.
- Implemented a two-stage LLM orchestration pipeline with query guardrails, delivered via a production FastAPI + React application with real-time streaming via Server-Sent Events.

End-to-End ASR Optimization | [GitHub](#) | Whisper, SpecAugment, MLflow, Docker

- Achieved > 48% absolute Word Error Rate (WER) reduction on the FLEURS benchmark by fine-tuning Whisper-small with SpecAugment, Label Smoothing, and Cosine LR scheduling.
- Reduced transcription errors across 177,000 unique terms by building a Devanagari normalization engine and fusing 5 distinct ASR models via majority-voting.

ACHIEVEMENTS & CERTIFICATIONS

- Ranked in the Top 1% globally (Gold Level) among thousands of participants in the WorldQuant BRAIN Global Alphathon 2022.
- Earned Kaggle 2× Expert status; solved 3,000+ algorithmic problems across LeetCode (Knight tier) and CodeChef (Max Rating 1662).
- Completed DeepLearning.AI Machine Learning & Deep Learning Specializations and earned the TensorFlow Developer Certificate.

SKILLS

Languages & Backend:	Python, SQL, C++, FastAPI, PostgreSQL, Server-Sent Events
AI & Machine Learning:	PyTorch, TensorFlow, LangChain, RAG, Prompt Engineering, LLM Security, Hugging Face, MPC
DevOps & Infrastructure:	AWS (EC2, SageMaker), Docker, CI/CD, Airflow, MLflow, Qdrant, Chroma